

AYUSH DODAL

AI ENGINEER

Mumbai, Maharashtra

+91 98195 60333

ayushdodal1999@gmail.com

linkedin.com/in/ayush-dodal-b5646016b/

PROFILE SUMMARY

AI Engineer with hands-on experience designing and deploying enterprise Generative AI, RAG, AI-agent, REST API, and data engineering solutions. Strong in Python, SQL, LangChain, LangGraph, FastAPI, vector databases, Azure, and PostgreSQL, with experience translating business workflows into scalable AI applications. Experienced in Agile delivery, document intelligence, data pipelines, prompt/RAG optimization, and production-oriented backend architecture.

WORK EXPERIENCE

Senior Research Fellow (AI Engineer)

June 2026 - Present

University of Mumbai | Mumbai, India

- **Production-Grade LLM Infrastructure:** Architecting a scalable FastAPI backend for automated task execution, implementing critical AI governance protocols—including structured schema validation, dynamic context budgeting, prompt versioning, and fallback logic—to ensure reliable, hallucination-free model outputs.
- **[Active Development] Multi-Agent Workflows:** Upgrading the core pipeline into an asynchronous, multi-agent framework using **LangGraph**, separating concerns into Decomposer, Executor, and Reviewer agents to increase task execution success rates and reliability.
- **[Active Development] AI Observability:** Integrating **Langfuse** for comprehensive LLM observability to track token usage, cost-per-run, and step-by-step latency, enabling data-driven prompt optimization and automated LLM-as-a-judge evaluation.

Database & AI Engineer

Nov 2025 - Present

American AI Tech | Mumbai, India

- **LLM Integration & RAG:** Developed AI-powered enterprise chatbots using **Retrieval-Augmented Generation (RAG)**, OpenAI embeddings, and ChromaDB, improving HR triage resolution rates by **40%** and drastically reducing manual intervention.
- **Backend API Architecture:** Designed scalable **FastAPI-based** RESTful APIs supporting high-volume AI workflows and document ingestion, consistently maintaining **sub-200ms latency** under enterprise load.
- **Data Ingestion Pipelines:** Built scalable ETL document ingestion workflows using Azure Blob Storage, successfully processing and vectorizing over **50,000+ enterprise documents** for accurate LLM retrieval.
- **Database Optimization:** Built secure PostgreSQL database architectures utilizing UUID-based relational modeling, reducing complex data retrieval times by **35%** to support fast AI operations.
- **Backend Automation:** Implemented PL/pgSQL triggers for foreign key resolution and conditional data processing, automating workflows and eliminating **15+ hours** of manual database administration per week.
- **Cross-Functional Collaboration:** Partnered with frontend teams in 2-week Agile sprints, utilizing Git-based version control to push **20+ production-ready** AI features and API endpoints to deployment.

TECHNICAL SKILLS

- **Programming:** Python, SQL, FastAPI, PL/pgSQL
- **Generative AI:** LLMs, RAG, Prompt Engineering, AI Agents, LangChain, ChromaDB, Vector Databases, Multi-Agent Workflows, LangGraph
- **Cloud & Big Data:** Azure Data Factory, Azure Databricks, Azure Blob Storage, Supabase
- **Databases:** SQL Server, PostgreSQL
- **Tools & Platforms:** Git, Azure DevOps, Selenium, ChromaDB
- **Concepts:** RESTful APIs, SDLC, Agile/Scrum, CI/CD, Row Level Security (RLS), Data Validation, Scalable System Design, Database Design

EDUCATION

Master of Science, Data Architecture & Management

2022 – 2024

Northeastern University, Boston

Bachelor of Technology, Computer Engineering

2018 – 2022

NMIMS University, Mumbai